SECTION 3: INPUT-OUTPUT RELATIONSHIP

3.1 THE LAW OF DIMINISHING RETURNS

Multiple Choice Questions

1990/CE/II/06

No. of Workers	Marginal Output
1	10
2	12
3	14
4	13
5	12

From the above table, the average output will be the highest when _____ workers are employed.

A. 2
B. 3
C. 4
D. 5

1992/CE/II/17
Suppose labour is the only variable factor in producing Good X. The daily total product of X is shown below:

Number of Workers	Total Product (units)
1	10
2	23
3	42
4	62
5	80

Diminishing marginal returns will set in after _____ workers are employed

A. 2
B. 3
C. 4
D. 5

1994/CE/II/21
The law of diminishing returns

A. states that firms will keep on adding variable factors to fixed factors
B. implies that firms will eventually operate at a loss.
C. implies that marginal product can be negative
D. explains production in the long run.

1996/CE/II/22
Short run is a period in which

(1) there are fixed factors of production.
(2) there are variable factors of production.
(3) the technology of production is fixed.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1) , (2) and (3)

1996/CE/II/23

No. of workers	1	2	3	4
Average product (units)	2	6	7	4

Refer to the above table. Diminishing marginal return sets in after ______ workers are employed in the production.

A. 1
B. 2
C. 3
D. 4

1999/CE/II/15

No. of workers	Marginal Output (Units)
1	10
2	12
3	11
4	10
5	7

From the above table, the average output will start to diminish when the _____ worker is put to work.

A. first
B. second
C. third
D. fourth

2002/CE/II/13
The following table shows the monthly production of a firm with two inputs, labour and machines.

Machines (Units)	Labour (Units)	Total Product (Units)
1	3	18
1	4	28
1	5	40
1	6	50
1	7	X

If the law of diminishing marginal returns applies to the firm, X

A. must be smaller than 50.
B. can be any value greater than 50.
C. must be smaller than 60.
D. is equal to 60.

2003/CE/II/20
The information below refers to Mr Chan's factory.

No. of workers	Average Output (Units)
10	5
11	6
12	7

Suppose diminishing marginal return sets in when more than 12 workers take part in production. The total output of 13 workers

A. can be greater than 102 units.
B. may be equal to 102 units.
C. must be less than 102 units.
D. is indeterminate.

2004/CE/II/18
The input-output relationship of Firm A is as follows:

Labour (Units)	Machinery (Units)	Average Output of Machinery (Units)
4	1	6
4	2	7
4	3	8
4	4	7
4	5	5

According to the above table, diminishing marginal returns set in when the ___ unit of machinery is employed.

A. 2nd
B. 3rd
C. 4th
D. 5th

2006/CE/II/17
The input-output relationship of Firm A is as follows:

Labour (Units)	Machine (Units)	Marginal Product of Labour (Units)
1	2	14
2	2	16
3	2	18
4	2	X

Diminishing marginal returns set in after the 3rd unit of labour is employed. The average product of 4 units of labour can be _______ units of output.

A. 16
B. 16.5
C. 17
D. 17.5

2008/CE/II/16
The input-output relationship of a firm is as follows:

Labour (units)	Machinery (units)	Average product of labour (units)
1	2	14
2	2	15
3	2	16

Diminishing marginal returns set in after the 3rd unit of labour is employed. The marginal product of the 4th unit of labour must be any number of output smaller than ________ units.

A. 15
B. 16
C. 17
D. 18

2010/CE/II/19
The input-output relationship of a firm is as follows:

Machine (units)	Labour (units)	Average product of labour (units)
3	1	20
3	2	12
3	3	11
3	4	10

According to the above table, diminishing marginal returns set in after _____ unit(s) of labour is/are employed.

A. 1
B. 2
C. 3
D. 4

2014/DSE/I/05
The following table shows a production plan of a firm which employs two factors only, namely labour and machinery.

Labour (unit)	Machinery (unit)	Total Product (unit)
1	4	100
2	4	120
3	4	150
4	4	170
5	4	165

The marginal product of labour will start to diminish when the _______ unit of labour is put to work.

A. second
B. third
C. fourth
D. fifth

2015/DSE/I/09
The table below shows the total product of Firm A, which employs only capital and labour in production.

Labour (units)	Capital (units)	Total Product (units)
12	2	110
12	3	130
12	4	160
12	5	X

Which of the following statements about Firm A is correct?

A. If the above data illustrate the law of diminishing marginal return, the value of X can be equal to 180.
B. If the above data illustrate the law of diminishing marginal return, the value of X can be equal to 190.
C. If the above data illustrate the law of diminishing marginal return, the value of X can be larger than 200.
D. The law of diminishing marginal return does not apply to Firm A because capital is not a fixed factor.

2018/DSE/I/10
The following table shows the input-output relationship of a firm . The only inputs required are capital and labour.

Capital (units)	Labour (units)	Marginal product of labour (units)
2	1	5
2	2	7
2	3	9
2	4	8
2	5	6

The average product of labour is at the highest when the quantity of labour is _______ units.

A. 2
B. 3
C. 4
D. 5

2025/DSE/I/06

The table below shows the input-output relationship of a firm. The only input required for the firm's production is labour. The wage rate per unit of labour is $50.

Labour (units)	Total output (units)
1	20
2	50
3	90
4	100
5	105

Based on the above information, we can conclude that_____,

A. the law of diminishing marginal returns applies to the firm because when the 4th unit of labour is employed, marginal product starts to diminish
B. when output increases from 90 units to 100 units, the firm suffers from diseconomies of scale
C. when output increases from 100 units to 105 units, the firm suffers a loss
D. the profit-maximising output is 90 units

Short & Structured Questions

1994/CE/I/11(a)
Mr Chan owns a farm. The following table shows the input-output relationship of this farm.

Fertilizers (Units)	Land (Units)	Labour (Units)	Total Output of Vegetables (Units)
0	10	40	30
1	10	40	70
2	10	40	120
3	10	40	160
4	10	40	190

State the law of diminishing returns and explain whether this law applies to Mr Chan's farm. (6 marks)

1997/CE/I/3
The following table shows the fall in total output of a factory when there are sick workers who cannot come to work.

No. of sick workers absent from work	Decrease in daily output (units)
1	20
2	50
3	90
4	140
5	200

Use the above data to illustrate the law of diminishing marginal returns. (5 marks)

1998/CE/I/4
The following table shows the weekly production of a garment factory with only two factors of production

No. of Machines	No. of Workers	Total Product (Units)
1	2	20
2	3	28
3	4	31
4	5	31
5	6	30

(a) State the law of diminishing marginal returns. (4 marks)

(b) Explain whether the above data illustrate the law of diminishing marginal returns. (2 marks)

2000/CE/I/4
The table below shows the production of a firm.

Machinery (units)	Labour (units)	Average output of labour (units)
10	7	30
10	8	34
10	9	37
10	10	39
10	11	40

State the law of diminishing marginal returns and explain whether the above data illustrate the law. (6 marks)

2001/CE/I/3
The table below shows the production of a firm.

Labour (Units)	Machinery (Units)	Total Product (Units)
10	5	100
11	5	150
12	5	250
13	5	400
14	5	530

Explain which law in economics can be illustrated by the above data. (7 marks)

2003/CE/I/2
(a) State the law of diminishing marginal returns. (3 marks)

(b) The following table shows the input-output relationship of a firm. Explain why the table illustrates the law in (a).

Machine (Unit)	Labour (Unit)	Average output of machine (Unit)
1	3	12
2	3	13
3	3	14
4	3	13

(3 marks)

2006/CE/I/4
The table below shows the input-output relationship of a firm.

Labour (Units)	Machine (Units)	Total output of labour (Units)
2	3	40
3	3	90
4	3	150
5	3	180
6	3	200

State the law of diminishing marginal returns and explain whether the above data illustrate the law. (6 marks)

2007/CE/I/3
(a) State the law of diminishing marginal returns. (3 marks)

(b) Study the following information on a factory.

Machines (units)	Labour (units)	Average product (units)
1	5	40
1	6	70
1	7	90
1	8	100

Explain whether the above data illustrate the law of diminishing marginal returns. (4 marks)

2009/CE/I/3

The table below shows the input-output relationship of a firm.

Labour (Units)	Machine (Units)	Average output of labour (Units)
1	2	10
2	2	12
3	2	14
4	2	16
5	2	17

State the law of diminishing marginal returns and explain whether the above data illustrate the law. (6 marks)

2012/DSE/II/2

The table below shows the input-output relationship of a firm.

Workers (units)	Machine (units)	Total output (units)
10	1	10
15	2	23
20	3	42
25	4	62
30	5	80

State the law of diminishing marginal returns and explain whether the above data illustrate the law. (5 marks)

2016/DSE/II/3

The table below shows the input-output relationship of a firm .

Machine (units)	Labour (units)	Average output of labour (units)
2	1	10
2	2	12
2	3	14
2	4	14
2	5	13

State the law of diminishing marginal returns and explain whether the above data illustrate the law. (6 marks)

2021/DSE/II/1c

The following is the advertisement of an online food delivery firm for recruiting riders.

<table>
<tr><td colspan="2">Riders Wanted !!!
</td></tr>
<tr><td><u>Earn great money</u>
• High salary for each delivery!
• Keep 100% tips from customers</td><td><u>Bring your own equipment</u>
• A smartphone
• A motorcycle with the necessary safety equipment</td></tr>
</table>

(c) After employing more riders, the firm found that its system was overloaded and the marginal product of riders dropped. State the law that explains this phenomenon and briefly explain why such law can be applied in this case. (5 marks)

2023/DSE/II/3

The table below shows the production data of a firm.

Machinery (units)	Labour (units)	Average output of labour (units)
12	10	10
12	11	12
12	12	15
12	13	17
12	14	16

(a) Based on the above data, explain why machinery is a fixed factor. (1 mark)

(b) Explain whether the law of diminishing marginal returns is illustrated by the above data. (3 marks)

2025/DSE/II/1

The table below shows the input-output relationship of a firm.

Machine (units)	Labour (units)	Average output of labour (units)
3	1	20
3	2	22
3	3	23
3	4	23
3	5	18

State the law of diminishing marginal returns and explain whether the above data illustrate the law. (5 marks)